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References

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*.
- Amos Azaria and Tom Mitchell. 2023. The internal state of an llm knows when it’s lying. *arXiv preprint arXiv:2304.13734*.
- Emily M Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*, pages 610–623.
- Lennart B rger, Fred A Hamprecht, and Boaz Nadler. 2024. Truth is universal: Robust detection of lies in llms. *arXiv preprint arXiv:2407.12831*.
- Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. 2022. Discovering latent knowledge in language models without supervision. *arXiv preprint arXiv:2212.03827*.
- Paul Christiano, Ajeya Cotra, and Mark Xu. 2021. Eliciting latent knowledge: How to tell if your eyes deceive you. URL https://docs.google.com/document/d/1WwsnJQstPq91_Yh-Ch2XRL8H_EpsnjrC1dwZXR37PC8/edit#heading=h.jrzi4atzacns.
- Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. 2019. *BoolQ: Exploring the surprising difficulty of natural yes/no questions*. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 2924–2936, Minneapolis, Minnesota. Association for Computational Linguistics.
- Corinna Cortes and Vladimir Vapnik. 1995. Support-vector networks. *Machine learning*, 20(3):273–297.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.
- Owain Evans, Owen Cotton-Barratt, Lukas Finnveden, Adam Bales, Avital Balwit, Peter Wills, Luca Righetti, and William Saunders. 2021. Truthful ai: Developing and governing ai that does not lie. *arXiv preprint arXiv:2110.06674*.
- Danny Halawi, Jean-Stanislas Denain, and Jacob Steinhardt. 2023. Overthinking the truth: Understanding how language models process false demonstrations. *arXiv preprint arXiv:2307.09476*.
- Jinwen He, Yujia Gong, Zijin Lin, Yue Zhao, Kai Chen, et al. 2024. Llm factoscope: Uncovering llms’ factual discernment through measuring inner states. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 10218–10230.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2020. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38.
- Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. 2023. Mistral 7b. *arXiv preprint arXiv:2310.06825*.
- Mandar Joshi, Eunsol Choi, Daniel S Weld, and Luke Zettlemoyer. 2017. Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1601–1611.
- Nitish Joshi, Javier Rando, Abulhair Saparov, Najoung Kim, and He He. 2024. Personas as a way to model truthfulness in language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 6346–6359.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. 2022. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*.
- Diederik P Kingma. 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*.
- Abhinav Kumar, Chenhao Tan, and Amit Sharma. 2022. Probing classifiers are unreliable for concept removal and detection. *Advances in Neural Information Processing Systems*, 35:17994–18008.
- Bruce W Lee, Benedict Florance Arockiaraj, and Helen Jin. 2023. Linguistic properties of truthful response. In *Proceedings of the 3rd Workshop on Trustworthy Natural Language Processing (TrustNLP 2023)*, pages 135–140.

- Benjamin A Levinstein and Daniel A Herrmann. 2024. Still no lie detector for language models: Probing empirical and conceptual roadblocks. *Philosophical Studies*, pages 1–27.
- Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. 2023. Inference-time intervention: Eliciting truthful answers from a language model. *Advances in Neural Information Processing Systems*, 36:41451–41530.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252.
- Dong C Liu and Jorge Nocedal. 1989. On the limited memory bfgs method for large scale optimization. *Mathematical programming*, 45(1):503–528.
- Kevin Liu, Stephen Casper, Dylan Hadfield-Menell, and Jacob Andreas. 2023. Cognitive dissonance: Why do language model outputs disagree with internal representations of truthfulness? In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 4791–4797.
- Monte MacDiarmid, Timothy Maxwell, Nicholas Schiefer, Jesse Mu, Jared Kaplan, David Duvenaud, Sam Bowman, Alex Tamkin, Ethan Perez, Mrinank Sharma, Carson Denison, and Evan Hubinger. 2024. Simple probes can catch sleeper agents.
- Matéo Mahaut, Laura Aina, Paula Czarnowska, Momchil Hardalov, Thomas Müller, and Lluís Marquez. 2024. Factual confidence of LLMs: on reliability and robustness of current estimators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4554–4570, Bangkok, Thailand. Association for Computational Linguistics.
- Alex Mallen and Nora Belrose. 2023. Eliciting latent knowledge from quirky language models. *arXiv preprint arXiv:2312.01037*.
- Samuel Marks and Max Tegmark. 2023. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*.
- Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. 2020. On faithfulness and factuality in abstractive summarization. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 1906–1919, Online. Association for Computational Linguistics.
- Shashi Narayan, Shay B. Cohen, and Mirella Lapata. 2018. Don’t give me the details, just the summary! topic-aware convolutional neural networks for extreme summarization. *ArXiv*, abs/1808.08745.
- Lorenzo Pacchiardi, Alex James Chan, Sören Mindermann, Ilan Moscovitz, Alexa Yue Pan, Yarin Gal, Owain Evans, and Jan M Brauner. 2023. How to catch an ai liar: Lie detection in black-box llms by asking unrelated questions. In *The Twelfth International Conference on Learning Representations*.
- F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830.
- J Platt. 1999. Probabilistic outputs for svms and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers*.
- CH-Wang Sky, Benjamin Van Durme, Jason Eisner, and Chris Kedzie. 2024. Do androids know they’re only dreaming of electric sheep? In *Findings of the Association for Computational Linguistics ACL 2024*, pages 4401–4420.
- Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.
- A Vaswani. 2017. Attention is all you need. *Advances in Neural Information Processing Systems*.
- Johannes Welbl, Nelson F Liu, and Matt Gardner. 2017. Crowdsourcing multiple choice science questions. *arXiv preprint arXiv:1707.06209*.
- Zhenhong Zhou, Haiyang Yu, Xinghua Zhang, Rongwu Xu, Fei Huang, and Yongbin Li. 2024. How alignment and jailbreak work: Explain LLM safety through intermediate hidden states. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 2461–2488, Miami, Florida, USA. Association for Computational Linguistics.

A Truth-related concepts

In this section, we distinguish the term “truthfulness” from several related concepts. Truthfulness refers to the alignment of a statement or (Q, A) with either world knowledge or contextual sources. The former, following Mahaut et al. (2024), is termed “factuality”. Contextual truthfulness, by contrast, may include fictional information that deviates from real-world facts, such as solving math problems in a hypothetical scenario.

Untruth lies at the negative end of the truthfulness spectrum and differs from *hallucination*, which refers to generations that are nonsensical or unfaithful to the provided source content (Ji et al.,

2023). A key distinction between untruth and hallucination is that truthfulness requires a sentence to be both sensible and unambiguous. Additionally, we differentiate untruth from *lies*. According to Pacchiardi et al. (2023), an answer is considered a lie only if the speaker knows the correct answer. In this view, a lie is a subset of untruths.

B Explanation on Choice of Probes

In this work we summarize LR, MLP, SVM and MM instantiations and use them for experiments. We do not use the TTPD probe introduced by the recent work, Bürger et al. (2024), for two reasons. First, the design of the TTPD probe is based on their finding that the “affirmative truth direction” and the “general truth direction” are not aligned. However, according to our findings in Section 4.3, the “affirmative truth direction” and the “general truth direction” become more consistent as the target model’s general capability increases. Therefore, for models with relatively high capability, the TTPD probe is not expected to distinguish itself among other probe instantiations.

Second, we find empirical evidence that the TTPD probe is not the most effective probe. We replicate the experiment on the BoolQ dataset in Section 4.6, and plot the output distribution and calibration curves of all probes targeting Llama-3.1-8B under the “with options” setting. The results are shown in Figure 9 and Figure 8. The output distribution of the TTPD probe resembles that of the MM probe, so does the calibration curve. This observation is not restricted to this setting and the BoolQ test set but is also noticed in a number of other settings and test sets. These findings suggest that judging by the functional behaviors of the TTPD probe, its performance is not representative enough to be reported.

C Details of Factual Datasets

We mentioned in Section 3.3 about the factual statements covering six topics which are used for training and testing truthfulness probes. We summarize each topic-specific dataset in Table 1, according to the data curators Azaria and Mitchell (2023), Marks and Tegmark (2023) and Bürger et al. (2024).

Instead of directly using the statements from Bürger et al. (2024), we perform minor modifications on the inventors topic and on logical disjunctions. We notice that the original inventors

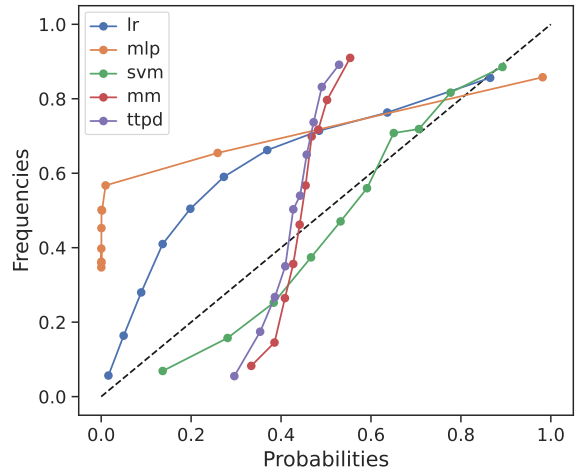


Figure 8: Calibration graph of LR, MLP, SVM, MM and TTPD probes on the BoolQ dataset under the “with options” setting. The target LLM is Llama-3.1-8B.

dataset has potential ambiguity due to duplication of name. Thus we specify that the person mentioned in a statement was an inventor, e.g. from “Thomas Edison lived in the U.S.” to “The inventor Thomas Edison lived in the U.S.”. Another tweak is that the original logical disjunctions composed by Bürger et al. (2024) are not consistent with conjunctions, as the subjects are written in full for conjunctions while the subjects are written in pronouns for disjunctions. To align these two logical transformations, we recover the subjects for disjunctions.

D Probe Implementation Details

Our implementation of the LR, SVM and MLP probes is based on the scikit-learn (Pedregosa et al., 2011) library. For the LR probe, we employ the L-BFGS optimization algorithm (Liu and Nocedal, 1989). For SVM, we utilize the NuSVC implementation. We set $\nu = 0.5$, a choice later validated by experiment results. Platt scaling is applied using five-fold cross-validation with the help of the scikit-learn library. For the MLP probe we configure a decreasing sequence of hidden units (512,128,64) with tanh activation, and we use the Adam optimizer (Kingma, 2014) to train it till convergence. Finally, we use the MM probe implementation provided by Marks and Tegmark (2023).

When establishing probes on atomic statements, we use a random 70% split for training and hold out the rest as the development set.